

NeighbourWise AI — Master Comprehensive Build Log

Project: DAMG 7374 — Group 4 | Northeastern University

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Coverage: Sessions 1–17 (February–April 2026) — All context logs, dev logs, Yash RAG checkpoints, Rutu individual work log, early pipeline documentation

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1. Project Overview & Vision

NeighbourWise AI is a Greater Boston neighborhood livability intelligence platform that answers natural language queries using a multi-agent AI architecture. It scores 51 neighborhoods across 9 livability domains and surfaces insights through a conversational Streamlit interface backed by Snowflake and Neo4j.

Core question answered: “Which neighborhood should I live in if I commute by transit, want low crime, need good schools, and care about restaurant variety?”

Project Timeline

Phase	Period	Focus
Phase 0 — Research	Feb 2–8, 2026	Dataset research, domain finalization, data profiling (Databricks)
Phase 1 — Pipelines	Feb 9–28, 2026	Airflow DAGs, S3, Snowflake STAGE tables, early dbt INT/MRT models
Phase 2 — Data Marts	Mar 1–15, 2026	All 9 domain MRTs, MASTER_LOCATION geometry, geocoding scripts
Phase 3 — Crime Science	Mar 7–12, 2026	DBSCAN + SARIMAX hotspot analysis, CA_ tables, crime Streamlit app
Phase 4 — RAG + Agents	Mar–Apr 2026	RAG pipeline unified, Graph Agent, Web Search Agent, Validator
Phase 5 — Application	Apr 2–17, 2026	FastAPI backend, Streamlit UI, guardrails, cost tracker, poster

Scale at a Glance

Metric	Value
Neighborhoods covered	51 (26 Boston + 13 Cambridge + 12 Greater Boston suburbs)
Livability domains	9 (Crime, Housing, Transit, Grocery, Healthcare, Schools, Restaurants, Universities, Bluebikes)
Crime records in Snowflake	356,852+ rows (Boston + Cambridge + Somerville)
Housing records	169,641 rows, 11 suburbs, 51 neighborhoods scored
RAG knowledge chunks	~8,400 chunks across 70 files (Rutu’s full corpus)
Airflow DAGs	7+ pipelines (weekly crime, annual MBTA/schools, quarterly restaurants)
AI agents deployed	6 (Graph, Router, Web Search, RAG, Report, Validator)
Report agent output	10-section magazine-edition PDF per

Metric	Value	
	neighborhood	

Domain Ownership

Domain	Owner	Notes
Crime / Safety	Aamir	DBSCAN + SARIMAX analysis
Housing	Rutu	169,641 rows, affordability scoring
Transit (MBTA + Bluebikes)	Aamir	3 sub-MRTs, tiered scoring v3
Grocery	Aamir	Geocoding + spatial joins
Healthcare	Rutu	DPH + OSM sources, DBSCAN, Streamlit dashboard
Schools	Aamir	MassGIS shapefile, geocoded
Restaurants	Rutu	Yelp Fusion + Harvard BARI reviews
Universities	Rutu	College Scorecard + Nominatim geocoding
RAG Pipeline	Yash + Rutu	neighbourwise_rag.py (Rutu built core script; Yash built original prototype)
Semantic Model	Rutu	20-table YAML, all 19 LOCATION_ID relationships
Agent Architecture	Rutu (initial design) + Aamir (LangGraph/Neo4j)	SQL+RAG+Orchestrator design

2. Infrastructure & Tech Stack

Layer	Technology	Details
Orchestration	Apache Airflow 2.8.1 (Docker)	CeleryExecutor, Redis broker, PostgreSQL metadata DB, port 8080
Storage	AWS S3	Bucket: neighborwise-ai-s3-bucket, region: us-east-2
Data Warehouse	Snowflake	Account: pgb87192 · Warehouse: NEIGHBOURWISE_AI · DB: NEIGHBOURWISE_DOMAINS · Role: TRAINING_ROLE
Transformation	dbt	Layers: STAGE →

Layer	Technology	Details
		INTERMEDIATE → MARTS → ANALYTICS
AI Generation	Claude claude-sonnet-4-6 (Anthropic)	Primary response generation for all natural language answers
AI Validation	GPT-4o (OpenAI)	Cross-model validator — independent of generation model
SQL + RAG	Snowflake Cortex / Mistral	mistral-large for narratives, e5-base-v2 for embeddings
Cortex Analyst	Snowflake Cortex Analyst	SQL Agent queries mart tables via semantic model YAML (Rutu)
Graph DB	Neo4j AuraDB	Gen-AI-Database instance for cross-domain graph queries
Agent Framework	LangGraph	Parallel fan-out for Graph Agent
Backend	FastAPI	uvicorn, port 8001, background tasks for report generation
Frontend	Streamlit	All data via FastAPI, configurable API_BASE_URL
Web Search	Serper API + Flask	Web Search Agent for real-time non-Snowflake queries
Image Gen	DALL-E 3 HD	4 perspectives per neighborhood, \$0.08/image
Geocoding	Nominatim (OpenStreetMap)	Free, no API key, 1.1s rate limit
OSM Extraction	Overpass API	Rutu's healthcare facilities extraction

Snowflake Schema Map

Schema	Table Pattern	Purpose
STAGE	STG_* tables	Raw data from Airflow DAGs (COPY INTO from S3)
INTERMEDIATE	INT_* tables	Cleaned, enriched, geocoded — dbt TABLE or INCREMENTAL
MARTS	MRT_* +	Domain-level analytics,

Schema	Table Pattern	Purpose
	MASTER_LOCATION	spatial joins, Cortex descriptions
ANALYTICS	NEIGHBORHOOD_MASTER_S CORE	Composite weighted score across all 9 domains
CRIME_ANALYSIS	CA_CRIME_*	DBSCAN clusters, SARIMAX forecast, Cortex safety narratives
GROCERY_ANALYSIS	GA_GROCERY_*	Grocery hotspot clusters
HEALTHCARE_ANALYSIS	HA_HEALTHCARE_*	Healthcare access profiles, hotspot clusters
RAW_UNSTRUCTURED	RAW_DOMAIN_CHUNKS	Vector embeddings for RAG (e5-base-v2, 768-dim)

CRITICAL: Snowflake account = pgb87192 (NOT pgb87192.sfedu02). Connector version must be snowflake-connector-python==3.6.0. Password is JWT OAuth token. All standalone geocoding scripts run on Mac, NOT inside Docker.

S3 Folder Structure

```
neighborwise-ai-s3-bucket/
├── crime-safety/
├── transit/
├── bluebikes/
├── schools/
├── housing/
├── restaurants/
├── grocery/
```

Environment Variables (Mac — standalone scripts)

```
SNOWFLAKE_ACCOUNT = pgb87192
SNOWFLAKE_USER    = BISON
SNOWFLAKE_PASSWORD = eyJ... (JWT token)
SNOWFLAKE_WAREHOUSE = NEIGHBOURWISE_AI
SNOWFLAKE_DATABASE = NEIGHBOURWISE_DOMAINS
SNOWFLAKE_SCHEMA   = STAGE
SNOWFLAKE_ROLE     = TRAINING_ROLE
S3_BUCKET          = neighborwise-ai-s3-bucket
```

3. Early Development Log — February 2026

Week of Feb 2–8: Research & Domain Finalization

- Researched datasets across crime, transit, bikes, schools, housing, restaurants, grocery, healthcare, universities domains
- Agreed on 8 core domains: Crime, Housing, Restaurants, Healthcare, Schools, Grocery, MBTA, Weather

- Performed data profiling in Databricks: row counts, null rates, schema, duplicate analysis, categorical frequencies
- Manually uploaded initial datasets to S3 for all domains except MBTA
- Finalized architecture: Data Sources → Airflow → S3 → Snowflake STAGE → dbt → INTERMEDIATE/MARTS

Week of Feb 16–17: Environment Setup + First Pipeline

Issue	Root Cause	Fix
Python 3.9 SIGSEGV on macOS Airflow	Gunicorn incompatibility with macOS	Switched to Docker-based Airflow 2.8.1
SSL certificate mismatch (Snowflake)	Educational account pgb87192.sfedu02 hostname mismatch	Direct connector with <code>insecure_mode=True</code>
Only 32,000 of 247,672 crime records fetched	Duplicate <code>limit=</code> parameters in API URL	Pagination while loop, removed hardcoded limit
CSV writing bug — only last batch saved	Using records instead of <code>all_records</code> variable	Changed to accumulated <code>all_records</code> list
Cannot create temp tables (Snowflake)	Missing schema context in session	Added <code>USE SCHEMA</code> before operations

Week of Feb 18–19: Additional Pipelines

MBTA Pipeline: Parallel fetch for routes, stops, and mapping. Direction-aware mapping (`DIRECTION_ID` 0=Outbound, 1=Inbound), stop sequencing, geographic coords and municipality.

Bluebikes Pipeline: Column mismatch between API response and table schema — fixed with column name logging.

Schools Pipeline: MassGIS shapefile (ZIP archive, DBF format). Files in ZIP subdirectory not found — fixed with `os.walk()` recursive search. Required `dbfread` library in Docker.

Week of Feb 20–28: Housing, dbt Setup, GitHub

Issue	Fix
Housing API 502 errors at high offsets	Retry logic with exponential backoff
Column count mismatch: CSV 67 cols, table 68 cols	Excluded auto-generated <code>LOAD_TIMESTAMP</code> from <code>COPY</code> column list
Comma-formatted numbers ('1,150') rejected by <code>NUMBER</code> type	Changed all housing numeric columns to <code>VARCHAR</code>
<code>MAIL_STATE</code> too small for 'SWITZERLAND'	Increased <code>VARCHAR</code> sizes
dbt double schema naming (<code>INTERMEDIATE_intermediate</code>)	Custom <code>generate_schema_name</code> macro
WKT strings too large for CSV upload	Geometry via direct Snowflake connector <code>UPDATE</code> in 5th DAG task

Rutu's dbt Environment Issues (Feb–Mar 2026)

- The .venv environment broke dbt due to a mashumaro/Python version incompatibility. Created and maintained a separate .venv-dbt (Python 3.12) exclusively for dbt commands
- VS Code auto-activates .venv on every new terminal — required: deactivate → `.\.venv-dbt\Scripts\Activate.ps1 → cd dbt\neighborhood_dbt` before any dbt work
- A broken universities sources.yml file blocked ALL dbt runs (even unrelated healthcare models) because dbt parse evaluates the entire project. Fixed the YAML to unblock the pipeline
- VS Code consistently saved new files to .venv/Scripts/ instead of scripts/ — had to use Save As (Ctrl+Shift+S) every time and verify with Get-Content | Select-String

Yash's Initial RAG Pipeline — Feb 28, 2026

First unstructured pipeline before the main RAG unification in March:

- **Schema:** RAW_UNSTRUCTURED. **Table:** RAW_PROXIMITY_GROCERY_CHUNKS — 3,802 chunks from 2 source files
- **Script:** snowflake_unstructured_loader.py — Hybrid chunker: Markdown heading-split first, char-window fallback. Min 150 chars
- Batch embedding: 50 chunks per UNION ALL. Model: e5-base-v2 with “passage:” prefix
- **Two-step INSERT:** temp VARCHAR table → INSERT INTO...SELECT PARSE_JSON()::VECTOR cast (Snowflake VECTOR type workaround)
- **Bugs fixed:** arctic-embed-m asymmetric scoring → reverted to e5-base-v2; short chunks ranking above content → 150-char minimum filter

4. MASTER_LOCATION — Geographic Backbone

MASTER_LOCATION is the central geographic reference table. Every domain mart produces exactly 51 rows via spatial join to this table. Lives in MARTS.MASTER_LOCATION, sourced from STAGE.STG_MASTER_LOCATION.

Final Composition (51 rows)

LOCATION_ID Range	Count	Flag	Cities/Neighborhoods
1–26	26	IS_BOSTON=TRUE	All 26 Boston neighborhoods
27–39	13	IS_CAMBRIDGE=TRUE	All 13 Cambridge neighborhoods
40–51	12	IS_GREATER_BOSTON=TRUE	Somerville(40), Arlington(41),

LOCATION_ID Range	Count	Flag	Cities/Neighborhoods
			Brookline(42), Newton(43), Watertown(44), Medford(45), Malden(46), Revere(47), Chelsea(48), Everett(49), Salem(50), Quincy(51)

Key columns: LOCATION_ID, NEIGHBORHOOD_NAME (always UPPERCASE), CITY, IS_BOSTON, IS_CAMBRIDGE, IS_GREATER_BOSTON, GRANULARITY ('NEIGHBORHOOD' for Boston/Cambridge, 'CITY' for Greater Boston suburbs), GEOMETRY (WKT polygon as GEOGRAPHY type), HAS_GEOMETRY (boolean), SQMILES, CENTROID_LAT, CENTROID_LNG, LOCATION_DESCRIPTION (Cortex via mistral-7b)

Spatial join pattern used throughout all dbt models:

ST_CONTAINS(m1.geometry, ST_MAKEPOINT(i.long, i.lat))

MASTER_LOCATION — [Change History](#)

Session	Date	Change	Outcome
Sessions 1-2	Feb 28	Initial 51 rows. Polygon geometry stored as GEOGRAPHY via ST_GEOGRAPHYFROMWKT. Cortex location_description generated.	Foundation established 
Session 3	Mar 6	CRITICAL: 12 Greater Boston cities had HAS_GEOMETRY=FALSE. Census TIGER API and Overpass API fixes both failed (504 timeouts). Fixed via fetch_greater_boston_wkt.py using GeoPandas + TIGER/Line 2023	All 12 cities have real polygons 

Session	Date	Change	Outcome
		shapefile (direct zip). Watertown = “Watertown Town” in TIGER — TIGER_NAME_OVER RIDES dict added.	
Session 3	Mar 6	Temporary expansion to 58 rows: 7 towns added (Winthrop, Waltham, Woburn, Winchester, Belmont, Milton, Weymouth) as LOCATION_IDs 52– 58	REVERTED — pending full domain coverage
Session 3	Mar 7	Cambridge SQMILES = -99 after full- refresh. Fix: UPDATE STG_MASTER_LOCATIO N SET SQMILES = ROUND(ST_AREA(...) /2589988.11, 2) WHERE IS_CAMBRIDGE = TRUE	Cambridge range: 0.24–1.23 sq mi 
Session 15	Apr 14	Geographic neighbors 404 for suburbs. AND GRANULARITY='NEIGH BORHOOD' excluded GRANULARITY='CIT Y' suburbs. Fixed to AND CENTROID_LAT IS NOT NULL	Suburbs resolve correctly 

5. Data Pipelines — Airflow DAGs

DAG	Source	Schedule	Records	Load Strategy
boston_crime_to _s3_to_snowflake	Analyze Boston API	Weekly Sun midnight	247,672+	MERGE on INCIDENT_NUM BER
cambridge_crim e_to_s3_to_snow flake	Cambridge Open Data	Weekly Sun midnight	108,726	TRUNCATE + reload

DAG	Source	Schedule	Records	Load Strategy
somerville_crime_to_s3_to_snowflake	Socrata API	Weekly Sun midnight	~22K	TRUNCATE + reload
greater_boston_crime_to_s3_to_snowflake	FBI CDE API	Monthly 1st 2am	—	MERGE (data through 2024 only)
mbta_api_to_s3_to_snowflake	MBTA API v3	Annually Jan 1	~500 route-stop rows	MERGE routes/stops, TRUNCATE mapping
bluebikes_api_to_s3_to_snowflake	Boston Open Data API	Quarterly	572 stations	MERGE on NUMBER
schools_to_s3_to_snowflake	MassGIS Shapefile (DBF)	Annually Jan 1	2,448	TRUNCATE + reload
housing_api_to_s3_to_snowflake	Boston Open Data API	Annually Jul 1	184,552	TRUNCATE + reload
yelp_restaurants_to_s3_to_snowflake	Yelp Fusion API	Quarterly	TBD	TRUNCATE + reload
STG_BOSTON_GROCERY	MassGIS/Infogr oup CSV	Manual	15,335	Manual upload
STG_BOSTON_HEALTHCARE	Massachusetts DPH	Airflow → S3 → Snowflake	626	TRUNCATE + reload
STG_BOSTON_HEALTHCARE_OSM	OpenStreetMap Overpass API	Manual/script	794	INSERT via executemany
master_location_dag	TIGER shapefile + hardcoded	Manual	51 rows	UPDATE for geometry

Standard DAG Pattern (all pipelines): 4 tasks: fetch_and_convert → upload_to_s3 → create_snowflake_table → load_to_snowflake. Resilient HTTP session: urllib3 Retry + per-page retry. TRUNCATE before COPY INTO prevents row duplication.

6. Geocoding Scripts

All geocoding scripts run standalone on Mac (NOT inside Docker). They use Nominatim (OpenStreetMap).

Common pattern: Nominatim rate limit 1.1s/request · Bounding box: lat 42.20–42.55, lng -71.35 to -70.85 · Sentinel value: -999.0 (NOT -99.0 which is a valid longitude) · Immediate

discard if outside bounding box · Checkpoint every N records → S3 → COPY INTO Snowflake

Script	Input Records	Success Rate	Runtime	Output Table	Status
geocode_schools.py	2,448	~100%	< 1hr	STG_BOSTON_SCHOOLS_GEOCODED	✓
geocode_grocery.py	15,335	—	~4.7 hrs	STG_BOSTON_GROCERY_GEOCODED	✓
geocode_housing.py	184,552	—	Multi-hour	STG_BOSTON_HOUSING_GEOCODED	✓
geocode_boston_crime.py	17,403	82.1% (14,283)	—	STG_BOSTON_CRIME_GEOCODED	✓
geocode_cambridge_crime.py	108,726	86.7% (94,250)	~10 hrs	STG_CAMBRIDGE_CRIME_GEOCODED	✓
geocode_universities.py (Rutu)	Boston area unis	—	—	STG_BOSTON_UNIVERSITIES_GEOCODED	✓

geocode_housing.py special optimizations: Three-layer: city pre-filter + ZIP pre-filter + address dedup (key: ST_NUM+ST_NAME+CITY). Bug fixed: NULL zip_code and NULL city caused AttributeError before .strip().

geocode_cambridge_crime.py scale: 3,556 actual Nominatim API calls, 104,872 cache reuses. Runtime: ~10 hours (01:14 → 11:18 Mar 8, 2026).

7. dbt Layer — STG / INT / MRT

CRITICAL Naming Convention

All Snowflake mart tables use **single-u NEIGHBORHOOD** (not double-u NEIGHBOURHOOD).

Correct Snowflake Table Name	Notes
MRT_NEIGHBORHOOD_SAFETY	Single-u
MRT_NEIGHBORHOOD_MBTA	Single-u — NOT MRT_NEIGHBORHOOD_TRANSIT

Correct Snowflake Table Name	Notes
MRT_NEIGHBORHOOD_BLUEBIKES	Single-u
MRT_NEIGHBORHOOD_SCHOOLS	Single-u
MRT_NEIGHBORHOOD_RESTAURANTS	Single-u
MRT_NEIGHBORHOOD_GROCERY_STORES	Single-u + _STORES suffix
MRT_NEIGHBORHOOD_HEALTHCARE	Single-u
MRT_NEIGHBORHOOD_UNIVERSITIES	Single-u
MRT_NEIGHBORHOOD_HOUSING	Single-u (teammate dbt ref uses double-u)
ANALYTICS.NEIGHBORHOOD_MASTER_SCORE	Composite — separate ANALYTICS schema

Teammate dbt source refs (exact spelling matters):

```

{{ source('marts', 'mrt_neighbourhood_housing') }}      -- double u
{{ source('marts', 'mrt_neighbourhood_grocery_stores') }} -- double u +
_stores
{{ source('marts', 'mrt_neighborhood_healthcare') }}     -- single u

```

INT Models

Model	Materialization	Key Derivations
INT_BOSTON_CRIME	Incremental (unique_key: incident_number)	time_of_day, is_weekend, season, crime_severity_tier, is_violent_crime, valid_location
INT_CAMBRIDGE_CRIME	Table	FILE_NUMBER PK, SPLIT_PART for date range format, Cambridge-specific severity terms
INT_SOMERVILLE_CRIME	Table	No lat/long — district-level only
INT_BOSTON_SCHOOLS	Table	INNER JOIN STG_BOSTON_SCHOOLS_GEOCODED on SCHID
INT_BOSTON_BLUEBIKE_STATIONS	Table	capacity_tier: LARGE ≥ 23, MEDIUM 15–22, SMALL 1–14
INT_BOSTON_RESTAURANTS	Table	LPAD(TRIM(ZIP_CODE), 5, '0') ZIP fix
INT_BOSTON_HOUSING	Table	Remove commas from VARCHAR numeric columns
INT_BOSTON_GROCERY	Table	INNER JOIN with STG_BOSTON_GROCERY_GEOCODED on SCHID

Model	Materialization	Key Derivations
int_boston_healthcare.sql (Rutu)	Table	OCODED Union DPH + OSM, ROW_NUMBER dedup (prefer DPH), geocoded join. 1,398 facilities
int_boston_restaurants.sql (Rutu)	Table	Cleaned, standardized, geocoded Yelp data
int_boston_universities.sql (Rutu)	Table	Joined with geocoded coords, polygon validation against MASTER_LOCATION

dbt Full Refresh Run Order

-- CASE-SENSITIVE model names; Cortex calls make safety/transit runs slow (~5-10 min)

```

dbt run --full-refresh --select INT_BOSTON_CRIME
dbt run --full-refresh --select INT_CAMBRIDGE_CRIME
dbt run --full-refresh --select INT_SOMERVILLE_CRIME
dbt run --full-refresh --select MRT_BOSTON_CRIME
dbt run --select MRT_NEIGHBORHOOD_SAFETY
dbt run --select MRT_NEIGHBORHOOD_MBTA
dbt run --select MRT_NEIGHBORHOOD_BLUEBIKES
dbt run --select MRT_NEIGHBORHOOD_SCHOOLS
dbt run --select MRT_NEIGHBORHOOD_RESTAURANTS
dbt run --select MRT_NEIGHBORHOOD_GROCERY_STORES
dbt run --select MRT_NEIGHBORHOOD_HEALTHCARE
dbt run --select MRT_NEIGHBORHOOD_UNIVERSITIES
dbt run --select MRT_NEIGHBORHOOD_HOUSING
dbt run --select NEIGHBORHOOD_MASTER_SCORE -- ALWAYS LAST

```

8. Crime Domain — Deep Dive

Crime Data Sources

Source	Stage Table	Rows	Notes
Analyze Boston API	STG_BOSTON_CRIME	247,672+	MERGE on INCIDENT_NUMBER, weekly
Cambridge Open Data	STG_CAMBRIDGE_CR IME	108,726	FILE_NUMBER PK, CRIME field, date range format
Somerville Socrata	STG_SOMERVILLE_C RIME	~22K	No lat/long
FBI CDE API	STG_GREATER_BOST	—	Data through 2024

Source	Stage Table	Rows	Notes
	ON_CRIME		only; next release ~Sep 2026

Cambridge vs Boston Crime Schema Differences

- FILE_NUMBER as PK (not INCIDENT_NUMBER). CRIME field (not OFFENSE_DESCRIPTION). LOCATION (not STREET)
- No OFFENSE_CODE (default -99), no DAY_OF_WEEK (derived), no SHOOTING (FALSE default)
- CRIME_DATE_TIME format: “MM/DD/YYYY HH:MM - HH:MM” → parsed with SPLIT_PART(..., ' - ', 1)
- Cambridge-specific severity: HOUSEBREAK (HIGH), MAL. DEST (MEDIUM), SHOPLIFTING (MEDIUM)

MRT_BOSTON_CRIME — Key Facts

- **Final row count: 356,852+** (Boston + Cambridge + Somerville unioned)
- COALESCE(ml.city, c.source_city) replaces hardcoded ‘BOSTON’ fallback
- ST_MAKEPOINT guard: CASE WHEN valid_location = TRUE THEN coord ELSE 0 END — prevents -999 coord Snowflake error

MRT_NEIGHBORHOOD_SAFETY — Score & Grade Evolution

Version	Problem	Change	Result
v1/v2	Max score 74.0; EXCELLENT threshold ≥ 75 permanently unreachable. 27/51 were GOOD.	—	Broken
v3 (Apr 14)	Thresholds recalibrated	≥ 65 EXCELLENT, ≥ 50 GOOD, ≥ 25 MODERATE, else HIGH CONCERN	EXCELLENT:10, GOOD:17, MODERATE:19, HC:4

Score range: 14.7 (Leather District) — 74.0 (South Boston Waterfront). 6-signal Z-score method; Greater Boston (FBI) uses 4 signals only.

9. All 9 Domain Marts

All marts produce exactly 51 rows (one per MASTER_LOCATION entry).

#	Mart Table	Score Method	Grade Labels	Notable
1	MRT_NEIGHBORHOOD_SAFETY	Z-score, 6 signals	EXCELLENT/GOOD/MODERATE	Range: 14.7–74.0

#	Mart Table	Score Method	Grade Labels	Notable
			/HIGH CONCERN	
2	MRT_NEIGHBORHOOD_MBTA	Route-based, tiered caps v3	EXCELLENT/GOOD/MODERATE/LIMITED	3 sub-MRTs. Score 7–97
3	MRT_NEIGHBORHOOD_BLUEBICKES	Additive: stations+docks+density	EXCELLENT/GOOD/MODERATE/LIMITED	Merged into Transit at 25% weight
4	MRT_NEIGHBORHOOD_SCHOOLS	Additive: count+mix+level+density	EXCELLENT/GOOD/MODERATE/LIMITED	501 rows in MRT_BOSTON_SCHOOLS
5	MRT_NEIGHBORHOOD_RESTAURANTS	Z-score, mean=50 std=15	EXCELLENT/GOOD/MODERATE/LIMITED	ZIP leading-zero fixed. Pho Viet dedup fixed
6	MRT_NEIGHBORHOOD_GROCERY_STORES	Additive: store types+density+pct essential	WELL_STOCKED/ADEQUATE/MODERATE/FOOD/DESERT	Table has _STORES suffix
7	MRT_NEIGHBORHOOD_HEALTHCARE	4-component composite: density+contact quality+diversity+capacity	EXCELLENT/GOOD/MODERATE/LIMITED + NO_DATA	DPH + OSM union. 1,398 facilities
8	MRT_NEIGHBORHOOD_UNIVERSITIES	Additive: presence+density+level+research+diversity+housing	EXCELLENT/GOOD/MODERATE/LIMITED/NONE	5 grades (others have 4)
9	MRT_NEIGHBORHOOD_HOUSING	Affordability-based (Rutu Shah)	AFFORDABLE/AVERAGE/BELLOW_AVERAGE	169,641 rows
10	ANALYTICS.NEIGHBORHOOD_MASTER_SCORE	Weighted composite of all 9	TOP PICK/SOLID CHOICE/MODERATE PICK/LIMITED APPEAL	FastAPI re-weights dynamically

Master Score Weights

Domain	Weight
Safety	20%

Domain	Weight
Transit (MBTA + Bluebikes combined)	20%
Housing	15%
Grocery	10%
Healthcare	10%
Schools	10%
Restaurants	10%
Universities	5%

Transit formula: $(mbta_score * 0.75) + (bikeshare_score * 0.25)$

Transit Score — v3 Tiered Caps

Problem (v1/v2): 4 neighborhoods flat at 100. 18 EXCELLENT, only 10 GOOD. Not discriminating.

Connectivity Tier	Cap
bus-only	70
commuter rail only	75
1 subway + ≤ 1 CR	78
1 subway + ≥ 2 CR	88
2 subway lines	92
3+ subway lines	97

Result: Score range 7–97. EXCELLENT:13, GOOD:15, MODERATE:12, LIMITED:11

MBTA Stop Override Fixes

- place-kenc1 (KENMORE) → BACK BAY (11): was attributing Green B/C/D to Fenway
 - place-bucen (BU CENTRAL) → BRIGHTON (18): was attributing Green B to Fenway
 - place-longw (LONGWOOD) → LONGWOOD (4): was bleeding into Brookline polygon
 - Stop 933 deduped via `ROW_NUMBER() OVER (PARTITION BY stop_id ORDER BY location_id ASC)` → Allston wins
-

10. Rutu Shah — Healthcare, Restaurants & Universities Domains

10.1 Healthcare Domain

Data Sources & Ingestion

- **Massachusetts DPH (Primary):** Licensed healthcare facilities with addresses, types, bed counts, phone numbers. 626 records into STG_BOSTON_HEALTHCARE via Airflow → S3 → Snowflake
- **OpenStreetMap Overpass API (Supplementary):** Rutu built `extract_osm_healthcare.py` to query hospitals, clinics, pharmacies, doctors, dentists, urgent care within the Greater Boston bounding box (42.20, -71.35 to 42.55, -70.85). 794 additional facilities into STG_BOSTON_HEALTHCARE_OSM
- **Health of Boston 2024 PDF:** Community health profiles, life expectancy, health equity data — loaded into RAG corpus
- **Boston Children's CHNA 2025:** Community Health Needs Assessment — qualitative health data for RAG
- **Massachusetts CMR Regulations (105 CMR 130, 140, 164):** Hospital licensure, clinic standards, long-term care regulations — loaded as policy PDFs for RAG

dbt Transformation Pipeline

Stage Layer:

Table	Records	Notes
STG_BOSTON_HEALTHCARE	626	Raw DPH facility data
STG_BOSTON_HEALTHCARE_GEOCODED	626	Lat/long from Nominatim geocoding
STG_BOSTON_HEALTHCARE_OSM	794	OpenStreetMap facilities

Intermediate Layer: `int_boston_healthcare.sql` — unions DPH and OSM datasets, deduplicates using `ROW_NUMBER()` (preferring DPH records over OSM for same facility), adds `source_system` column, joins geocoded coordinates. **Final: 1,398 facilities.**

Mart Layer — Row-Level: `MRT_BOSTON_HEALTHCARE` — one row per facility with: `FACILITY_ID`, `FACILITY_NAME`, `FACILITY_TYPE`, `FACILITY_TYPE_GROUP`, `LAT`, `LONG`, `NEIGHBORHOOD_NAME`, `IS_HOSPITAL`, `IS_CLINIC`, `HAS_VALID_PHONE`, `BED_COUNT`, `HAS_VALID_LOCATION`, `LOCATION_ID`, `SOURCE_SYSTEM`.

Mart Layer — Neighborhood Aggregated: `MRT_NEIGHBORHOOD_HEALTHCARE` — one row per neighborhood (all 51) with: `TOTAL_FACILITIES`, `HOSPITAL_COUNT`, `CLINIC_COUNT`, `TOTAL_BED_CAPACITY`, `FACILITIES_PER_SQ_MILE`, `HEALTHCARE_SCORE` (0–100), `HEALTHCARE_GRADE` (A–F + NO_DATA), Cortex-generated narrative.

The 51-Neighborhood Problem & Fix

Problem: Initial aggregated mart only showed 40 of 51 neighborhoods because: (1) SQL used `HAVING COUNT(h.facility_id) > 0`, which excluded zero-facility neighborhoods; (2) join was an `INNER JOIN` against `MASTER_LOCATION`.

Phase 1 (SQL fix): Removed `HAVING` clause, switched to `LEFT JOIN` from `MASTER_LOCATION`, wrapped all metrics in `COALESCE(..., 0)`, added `NO_DATA` grade for zero-facility neighborhoods. Added conditional Cortex `COMPLETE` call that skips narrative generation for empty neighborhoods to prevent hallucinated content.

Phase 2 (Data enrichment): Built `extract_osm_healthcare.py` for Overpass API to pull 794 OSM facilities. After deduplication in intermediate model, this filled 7 of 10 previously empty neighborhoods with real facility data. Three neighborhoods legitimately remain at zero: Harbor Islands, Bay Village, Strawberry Hill.

Healthcare DBSCAN Spatial Analysis

The existing “DBSCAN” was actually SQL grid binning (`ROUND(lat, 2)`, `ROUND(long, 2)`) — no real spatial clustering. Rutu replaced this with proper Python DBSCAN:

- Haversine distance metric, $\epsilon = 250\text{m}$ radius, `min_samples = 3` (wider than grocery — healthcare facilities are sparser)
- Per-neighborhood output: total facilities, type breakdown, cluster count, clustered share, isolated percentage, access tier classification
- Cortex-generated narratives per neighborhood
- Results written to `HEALTHCARE_ANALYSIS` schema in Snowflake

Healthcare Dashboard (Streamlit) — 7 Visualizations

1. 5 bordered KPI cards: total facilities, hospitals, clinics, avg score, coverage rate
2. Lollipop chart for neighborhood score rankings
3. Choropleth map with DBSCAN facility scatter overlay (red dots = hospitals, green = clustered, grey = isolated) using pydeck
4. Access tier donut chart
5. Score vs. density bubble scatter
6. Score component heatmap
7. Facility type mix (100% stacked bar chart)

Healthcare Scoring Model

4-component composite (0–100): facility density + contact quality (phone availability) + type diversity (mix of hospitals, clinics, pharmacies) + bed capacity. Each component normalized and weighted. Neighborhoods graded A–F + `NO_DATA`. Every neighborhood gets an LLM-generated narrative translating the score into actionable insight.

10.2 Restaurants Domain

Data Sources

- **Yelp Fusion API (Primary):** Restaurant listings with ratings, review counts, price tiers, cuisine categories, delivery/pickup options. The Yelp API enforces a **1,000-record cap per query** — overcome using geographic chunking combined with category-based extraction
- **Harvard BARI Yelp Reviews:** Pre-cleaned review dataset from Harvard Dataverse for the Boston area — used for unstructured RAG content

dbt Models

Model	Contents
STG_BOSTON_RESTAURANTS	Raw Yelp API data
int_boston_restaurants.sql	Cleaned, standardized, geocoded
MRT_BOSTON_RESTAURANTS	Individual restaurants: ratings, reviews, price tiers, cuisine types, delivery options, neighborhood assignment
MRT_NEIGHBORHOOD_RESTAURANTS	Aggregated: restaurant density, avg rating, cuisine diversity index, price distribution

Unstructured Data (RAG)

Restaurant-related policy PDFs: Boston food establishment inspection regulations, Massachusetts sanitary code (105 CMR 590), food truck/mobile vendor regulations, food safety grading standards. **~1,286 chunks** in the RESTAURANTS domain in RAW_DOMAIN_CHUNKS.

10.3 Universities Domain

Data Sources

- **US Department of Education College Scorecard:** Program descriptions, institutional characteristics, enrollment, admission rates, narrative fields for every accredited institution in the Boston area
- **Nominatim Geocoding:** Bounding box + polygon validation against MASTER_LOCATION to assign each university to its correct neighborhood

dbt Models

Model	Contents
STG_BOSTON_UNIVERSITIES	Raw College Scorecard data
STG_BOSTON_UNIVERSITIES_GEOCODED	Lat/long from Nominatim geocoding
int_boston_universities.sql	Joined with geocoded coords, cleaned and standardized
MRT_NEIGHBORHOOD_UNIVERSITIES	Aggregated: EDUCATION_SCORE, 5 grades

Model	Contents
	(EXCELLENT/GOOD/MODERATE/LIMITED/NONE)

Unstructured Data (RAG)

University policy and report PDFs: enrollment reports, student housing impact assessments, economic impact studies. ~**158 chunks** in the UNIVERSITIES domain in RAW_DOMAIN_CHUNKS.

10.4 Semantic Model YAML (Cross-Domain)

Rutu authored `scripts/neighbourwise_semantic_model.yaml` — the contract between the Cortex Analyst SQL Agent and the Snowflake mart layer. Defines all **20 mart tables** across all teammates' domains with their dimensions, measures, and **19 LOCATION_ID-based relationships**. This tells the LLM what tables exist, what columns mean, and how tables relate.

Uploaded to Snowflake stage: `@NEIGHBOURWISE_DOMAINS.MARTS.AI_MODELS/` via `scripts/upload_yaml.py`.

10.5 Presentation & Deliverables (Rutu)

- System architecture diagram (high-res PNG, 2400×2380px) — all 9 layers from data sources to Streamlit frontend
- Data engineering architecture diagram
- 15-slide presentation deck (dark theme, matching team visual identity)
- One-page academic poster for DAMG7374
- Professional README for `github.com/NeighbourWise-AI/neighbourwise.ai_version1`
- `requirements.txt` for the project repository
- `overview_endpoints.py` — FastAPI endpoint definitions for all 9 domain dashboards

11. RAG Pipeline — Yash Khavnekar & Rutu Shah

Table: `NEIGHBOURWISE_DOMAINS.RAW_UNSTRUCTURED.RAW_DOMAIN_CHUNKS`

Two scripts: - `snowflake_unstructured_loader.py` — Yash's original Feb 2026 prototype (grocery domain proof of concept) - `scripts/neighbourwise_rag.py` — Rutu's production pipeline used by all teammates, with 3 CLI stages: download / load / search

RAG Pipeline Schema

chunk_id	AUTOINCREMENT PK
domain	VARCHAR
source_file	VARCHAR
chunk_index	NUMBER
chunk_text	VARCHAR
chunk_embedding	VECTOR(FLOAT, 768)
loaded_at	TIMESTAMP_NTZ

Embedding: e5-base-v2 via Snowflake Cortex `EMBED_TEXT_768()`. Ingest prefix: `passage:`
Search prefix: `query:`

Hybrid scoring: 65% vector similarity + 35% keyword matching (Cortex Search Service: `NEIGHBOURWISE_SEARCH`)

How neighbourwise_rag.py Works (Rutu)

- **Download stage:** Fetches PDF from URL, saves locally, extracts raw text using pdfplumber with page markers (`--- Page N ---`). Organized into `rag_docs/<domain>/` folders
- **Load stage:** Chunks text into ~1,000-char pieces with 200-char overlap, embeds via Snowflake Cortex (e5-base-v2), bulk-inserts into `RAW_DOMAIN_CHUNKS` using `executemany + single conn.commit()`
- **Search stage:** Queries Cortex Search Service (`NEIGHBOURWISE_SEARCH`) with hybrid retrieval, returns ranked chunks with similarity scores

Critical RAG Fixes (Rutu)

Fix	Problem	Solution
embed_batch sanitization	Dense legal PDFs broke embed_batch due to backslashes and non-printable chars	Rewrote sanitization pipeline: replace backslashes first → escape single quotes → filter non-printable/null chars → truncate to 2,000 chars
Bulk insert	Row-by-row insert loop caused 336,000 duplicate rows in one incident	Switched to <code>executemany + single conn.commit()</code> pattern
Deduplication	Duplicate files across domain assignments (Live_Long_and_Well under both HEALTHCARE and GENERAL; Full_recomendations_0.txt under both RESTAURANTS and SCHOOLS)	Identified and resolved all duplicates in <code>RAW_DOMAIN_CHUNKS</code>

Policy PDF Curation (Rutu)

Curated and loaded **32 policy PDFs across all 8 domains**, including Massachusetts CMR regulations, Boston city policy documents, MBTA service policies, housing strategy documents, and school regulations. Several government sites (malegislature.gov, some mass.gov URLs) blocked automated downloads via SSL reset — downloaded manually via browser as fallback.

Final corpus: ~8,400 chunks across 70 files in RAW_DOMAIN_CHUNKS.

Chunk Status (Combined)

Domain	Chunks	Status	Owner
CRIME	~415	✓ Complete	Aamir
TRANSIT	~660	✓ Complete	Aamir
BLUEBIKES	~230	✓ Complete	Aamir
SCHOOLS	~130	✓ Complete	Aamir
UNIVERSITIES	~158	✓ Complete	Rutu
GENERAL	~200	✓ Complete	Rutu
RESTAURANTS	~1,286	✓ Complete	Rutu
HEALTHCARE	Loaded	✓ Complete	Rutu
HOUSING	Loaded	✓ Complete	Rutu

Full corpus: ~8,400 chunks / 70 files

RAG Cleanup Pipeline (in report_agent.py)

- `clean_rag_chunks()` — strips OCR artifacts (--- Page N ---, (cont.)), special chars, fragments <80 chars)
- `summarize_rag_with_cortex()` — sends cleaned top 3 chunks to Cortex with magazine-editorial prompt

12. Crime Hotspot Analysis — DBSCAN & SARIMAX

Script: `crime_hotspot_analysis.py` | **Schema:** CRIME_ANALYSIS

Tables Produced

Table	Description
CA_CRIME_FORECAST	SARIMAX 6-month forecast. Columns: LOWER_CI, UPPER_CI (NOT LOWER_BOUND/UPPER_BOUND)
CA_CRIME_HOTSPOT_CLUSTERS	Two-pass DBSCAN cluster-level stats
CA_CRIME_CLUSTER_POINTS	Individual crime points with cluster labels (for map viz)
CA_CRIME_SAFETY_NARRATIVE	Cortex mistral-large 2–3 sentence safety

Table	Description
	narrative, reliability flag HIGH/LOW

Configurations

SARIMAX: MIN_MONTHS=12, MIN_MONTHS_SEASONAL=24, FORECAST_HORIZON=6, D=1 (forced). Uses pmdarima auto_arima.

DBSCAN — Stats pass: EPS_KM=0.075 (75m), MIN_SAMPLES=20

DBSCAN — Map/viz pass: EPS_KM_MAP=0.150 (150m), MIN_SAMPLES_MAP=120

Critical Bug Fixed — DBSCAN Metric

Original: StandardScaler on lat/long then multiply eps by 111. This is **geometrically incorrect** — StandardScaler normalizes to unit variance, destroying the degree-to-distance relationship.

Fix: metric="haversine" on raw radians with $\text{eps} = \text{EPS_KM} / 6371.0$.

Known Issues

- Cambridgeport ARIMA: 41→5, MAPE=999 — flagged LOW reliability
- MAPE cap only in CA_CRIME_SAFETY_NARRATIVE, not CA_CRIME_FORECAST
- Harbor Islands leaks into CA_CRIME_FORECAST and CA_CRIME_SAFETY_NARRATIVE — HARBOR_FILTER not applied
- load_dotenv() fix (Session 14): added from dotenv import load_dotenv; load_dotenv() to avoid manual env export

13. Agent Architecture

Agent	Technology	Status	Key Detail
Graph Agent	LangGraph + Neo4j AuraDB	✅ Complete	Parallel fan-out. ~5s retrieval. 100/100 validator pass
Router Agent	Python + Claude API	✅ Complete	3-stage: classify → dispatch → normalise. RoutingMeta in response
Web Search Agent	Serper + Claude + GPT-4o	✅ Complete	95/100 on Crime and MBTA queries
RAG Agent	Snowflake Cortex + e5-base-v2	✅ Complete	Hybrid scoring, domain chunks
Report Agent v2.1	ReportLab + Altair + Matplotlib	✅ Complete	10-section magazine PDF. generate_report(ne

Agent	Technology	Status	Key Detail
Universal Validator	GPT-4o	✅ Complete	neighborhood) — manages own Snowflake conn 100/100 first-attempt pass after rank citation fix
Graphic Agent	Altair + Plotly + DALL-E 3 HD	✅ Complete	6 chart types + 4 AI images per neighborhood
Validator Agent (Rutu)	Claude 3.5 Sonnet via Cortex	✅ Complete	validate_and_improve() — hallucination, format, accuracy. Called on every query
Cortex SQL Agent (Rutu)	Cortex Analyst + Semantic Model YAML	✅ Complete	Queries Snowflake mart tables via 20-table semantic model

Rutu's Initial Three-Agent Architecture Design

Rutu designed and built the initial three-agent hybrid system before the LangGraph integration:

- **SQL Agent:** Cortex Analyst queries Snowflake mart tables via the semantic model YAML. Generates SQL, runs it, returns structured data
- **RAG Agent:** Cortex Search Service queries RAW_DOMAIN_CHUNKS via vector search. Retrieves relevant document chunks
- **Orchestrator:** Cortex COMPLETE with mistral-large2 classifies each query and routes to SQL, RAG, or BOTH. Synthesizes results into three-section format (Summary, Key Data, Insights)

Non-Deterministic Routing Fix (Rutu)

Problem: Ambiguous queries like “Is Allston safe and affordable?” were inconsistently routed to RAG-only vs. BOTH across separate runs, causing incomplete answers.

Root cause: `classify_question` used `cortex_complete` with `mistral-large2` at temperature `> 0` — variable classification on boundary queries.

Fix	Change
Fix 1	Set <code>temperature=0</code> for deterministic classification

Fix	Change
Fix 2	Updated classification prompt to bias toward BOTH for neighborhood-specific queries
Fix 3	Added keyword overrides in <code>ask_neighbourwise</code> — when data-heavy terms (safe, affordable, rent, hospitals, etc.) detected, routing forced to BOTH regardless of LLM classification

SQL Agent Issues Fixed (Rutu)

- SQL agent generated bare table names (not full paths) — enforced `NEIGHBOURWISE_DOMAINS.MARTS.TABLE_NAME` format
- Mixed-case neighborhood names — enforced UPPERCASE neighborhood names in all queries

Router Agent — Bugs Fixed (Aamir)

Bug	Root Cause	Fix
“Compare Back Bay and Roxbury across all domains” → <code>web_search</code>	<code>compare/vs/versus</code> not in chart keywords	Added to <code>_keyword_classify()</code>
<code>KeyError: intent</code> in classification prompt	Unescaped JSON braces with <code>.format()</code>	Doubled all braces <code>{{</code> and <code>}}</code>
HTTP calls to Graph Agent	<code>handle_graph_query()</code> making HTTP calls	Direct import: <code>from Graph_agent import ask_graph_agent</code>
0-domain queries hitting Snowflake	Cortex fallback before <code>web_search</code> dispatch	Skip Snowflake entirely for 0-domain queries

Report Agent v2.1 — 10 Report Sections

1. **Cover page** — name, city, hero score 64pt, grade, strength/weakness pills, date
2. **Executive summary** — Cortex mistral-large, 3 paragraphs, magazine-editorial tone
3. **Domain scorecard table** — 9 scores + grades, color-coded
4. **Radar chart** — Matplotlib 7x7, domain-colored dots
5. **Grouped bar chart** — Altair horizontal (9 domains), sorted ascending
6. **Neighborhood comparison** — Altair, blue accent for focus, 5 nearest neighbors
7. **Crime trend + SARIMAX forecast** — Altair area + dashed line + CI band + narrative
8. **Per-domain narratives** — from `_DESCRIPTION` columns
9. **Lifestyle context** — RAG chunks → `clean_rag_chunks()` → `summarize_rag_with_cortex()` → 2 paragraphs
10. **Recommendation** — Cortex, numbered points in blue-tinted pull-quote box

Validated: Fenway (6 pages)  | Roxbury (9 pages)  | Dorchester (SARIMAX overlay) 

14. FastAPI Backend

File: neighbourwise_fastapi.py / neighborwise_fastapi.py **Run:** python3 -m uvicorn neighbourwise_fastapi:app --reload --port 8001

All Endpoints (All Verified)

Method	Endpoint	Description
GET	/health	Health check
GET	/overview/neighborhoods	All 51 neighborhoods list
GET	/overview/kpis	KPI stats
GET	/overview/map	GeoJSON for pydeck choropleth
GET	/overview/crime-summary	Crime summary with monthly_by_city breakdown
GET	/overview/domain/{domain }	All 9 domains
GET	/overview/domain-matrix	All domains, all neighborhoods
POST	/query	NL query → Agent dispatch → Response
POST	/report/generate	Trigger background report generation
GET	/report/{report_id}	Poll report status
GET	/report/{report_id}/download	Download PDF bytes
GET	/report	List all reports

/query Endpoint — 4 Layers

1. **Input guardrails** — regex-based, blocks jailbreaks/toxic/off-topic before any DB work
2. **Cost tracker** — estimates tokens per call (classify → generate → validate → improve)
3. **Output guardrails** — PII redaction + hallucination detection
4. **llm_usage** returned for Streamlit cost tracker panel

Key FastAPI Fixes

Bug	Fix
ask_report_agent(neighborhood, conn)	generate_report(neighborhood) —

Bug	Fix
wrong signature	report_agent manages its own conn
generate_report name collision	Renamed endpoint to generate_report_endpoint
Old broken teammate endpoints returning 500	Removed /overview/stats, /overview/safety, /overview/hotspots, /overview/crime-narratives etc.
Grocery domain 500 error for specific neighborhoods	OPEN — column name mismatch

Column Name Gotchas

- MRT_NEIGHBORHOOD_GROCERY_STORES: single-u NEIGHBORHOOD in Snowflake
- MARTS.MASTER_LOCATION: GEOMETRY column (geography type) — **NOT** GEOMETRY_WKT
- CA_CRIME_FORECAST: LOWER_CI, UPPER_CI — **NOT** LOWER_BOUND/UPPER_BOUND
- All NEIGHBORHOOD_NAME values in Snowflake are **UPPERCASE**
- MASTER_LOCATION.GRANULARITY: Boston/Cambridge = 'NEIGHBORHOOD', Greater Boston = 'CITY'
- HARBOR_FILTER = "AND UPPER(NEIGHBORHOOD_NAME) != 'HARBOR ISLANDS'" — missing from CA_CRIME_FORECAST

15. Streamlit Frontend

File: neighbourwise_app.py | **Run:** streamlit run neighbourwise_app.py

Three Tab Structure

Overview Tab — Mode 1 (Home): NOT domain_filter AND NOT hood_filter 5 KPI cards · safety choropleth pydeck map · safety donut · affordability bar · transit line pills · livability heatmap · crime trend narrative

Overview Tab — Mode 2 (Domain Deep-Dive): domain_filter is set - Safety city-wide: 5 safety KPIs + 3 trend KPIs (colored red/yellow/green) + DBSCAN hotspot bar + monthly crime trend - Safety single-neighborhood: 5 metric cards + AI narrative + SARIMAX forecast + DBSCAN donut - Healthcare (Rutu): KPI cards, lollipop chart, choropleth + scatter overlay, access tier donut, bubble scatter, heatmap, stacked bar

Overview Tab — Mode 3 (Neighborhood Profile): hood_filter AND NOT domain_filter Hero card · domain scorecard bar (all 9) · 6 domain cards in 2-col layout · ranking vs all 50

Ask Tab: 6 example buttons · POST /query timeout=300s · Renders: answer, SQL expander, results table, RAG sources, routing metadata, intent badge

Report Tab: POST /report/generate → poll every 8s → st.download_button for PDF bytes

Streamlit Bugs Fixed

Bug	Fix
Sidebar disappeared	Browser localStorage caching. <code>localStorage.clear()</code> in DevTools + CSS <code>[data-testid="collapsedControl"] { display: none; }</code>
Cost tracker only fired for report queries	<code>_log_query_cost</code> inside <code>if rtype == "report":</code> block — moved outside
KPI cards show neighborhood data when <code>hood_filter</code> active	<code>load_kpis()</code> must not receive <code>hood_filter</code> (OPEN)
Safety donut shows 100% single grade with <code>hood_filter</code>	(OPEN)
Transit map black background	No Mapbox token — bar chart fallback
Git merge conflict during live session	<code>git config pull.rebase false; git pull origin main</code>

Session 17 Additions

- Cost tracker panel: `_esc()`, `_init_cost_tracker()`, `_log_query_cost()`, `_render_cost_tracker()` wired
- Guardrails: blocked query card (red box) in `render_assistant_message()` + PII/flagged captions
- Safety deep-dive fully rebuilt: city-wide + single-neighborhood views
- `monthly_by_city` field added to `/overview/crime-summary` endpoint
- Home KPI `render_metric_cards()` moved inside `else: block` (Mode 1 only)

16. Scoring Methodology

Domain	Method	Score Range	Grades
SAFETY	Z-score, 6 signals (4 for Greater Boston)	14.7–74.0	EXCELLENT / GOOD / MODERATE / HIGH CONCERN
TRANSIT (MBTA)	Route-based, tiered caps v3	7–97	EXCELLENT / GOOD / MODERATE / LIMITED
BLUEBIKES	Additive: stations + docks + density	—	EXCELLENT / GOOD / MODERATE / LIMITED
SCHOOLS	Additive: count + mix + level + density	—	EXCELLENT / GOOD / MODERATE / LIMITED
RESTAURANTS	Z-score, mean=50 std=15	—	EXCELLENT / GOOD / MODERATE /

Domain	Method	Score Range	Grades
GROCERY	Additive: store types + density + pct essential	—	LIMITED WELL_STOCKED / ADEQUATE / MODERATE / FOOD_DESERT
HEALTHCARE	4-component composite: density + contact quality + type diversity + bed capacity	0–100	A / B / C / D / F / NO_DATA
UNIVERSITIES	Additive: presence+density+level+research+diversity+housing	—	EXCELLENT / GOOD / MODERATE / LIMITED / NONE
HOUSING	Affordability-based (Rutu Shah)	—	AFFORDABLE / AVERAGE / BELOW_AVERAGE
MASTER	Weighted composite	—	TOP PICK / SOLID CHOICE / MODERATE PICK / LIMITED APPEAL

Cortex Prompt Standard — All Domains

- Model: mistral-large (upgraded from mistral-7b which hallucinated wrong states/cities)
- Always include ', Massachusetts' after city name
- Always include 'IMPORTANT: [neighborhood] is in [CITY], Massachusetts'
- Cortex + c.* causes “Object C does not exist” — always list columns explicitly
- SQL REGEXP_REPLACE post-processing: catches "city of X" → "the X neighborhood of Boston/Cambridge"
- Conditional skip for zero-facility neighborhoods to prevent hallucinated narratives (Rutu’s healthcare fix)

17. Issues & Bugs Master Log

Infrastructure / Data Layer

Session	Issue	Root Cause	Fix	Status
Feb 16	Python 3.9 SIGSEGV on macOS Airflow	Gunicorn incompatibility	Docker-based Airflow 2.8.1	✓
Feb 16	SSL cert	pgb87192.sfedu	insecure_mode=	✓

Session	Issue	Root Cause	Fix	Status
	mismatch (Snowflake edu account)	02 hostname mismatch	True	
Feb 18	Only 32K of 247K crime records fetched	Duplicate limit= in API URL	Pagination while loop	✓
Feb 18	CSV writing bug — only last batch saved	records vs all_records variable	Changed to accumulated list	✓
Feb 19	Schools shapefile not found in Docker	Files in ZIP subdirectory	os.walk() recursive search	✓
Feb 20	Housing API 502 errors at high offsets	Large dataset, unstable API	Retry logic with exponential backoff	✓
Feb 20	Housing column count mismatch (CSV:67, table:68)	Auto-generated LOAD_TIMESTAMP	Excluded from COPY column list	✓
Feb 20	Comma-formatted numbers ('1,150') rejected	NUMBER type	Changed all housing numeric cols to VARCHAR	✓
Feb 20	dbt double schema naming	Default dbt schema behavior	Custom generate_schema_name macro	✓
Feb 20	WKT strings too large for CSV	GEOGRAPHY type	Direct Snowflake connector UPDATE in 5th DAG task	✓
Feb 28	Yelp API rate limiting	Free tier	Reduce scope / cached data	⚠ Partial
Mar	Healthcare mart only showed 40/51 neighborhoods	INNER JOIN + HAVING COUNT > 0	LEFT JOIN + COALESCE + conditional Cortex skip	✓
Mar	OSM data 336K duplicate rows	Row-by-row insert loop	executemany + single conn.commit()	✓

Session	Issue	Root Cause	Fix	Status
Mar	OSM VARIANT type mismatch for RAW_TAGS	Column type incorrect	Changed to VARCHAR(16777216)	✓
Mar 6	12 Greater Boston cities HAS_GEOMETRY=FALSE	Not in original DAG	fetch_greater_boston_wkt.py via GeoPandas+TIGER	✓
Mar 6	Census TIGER/Overpass API fix attempts failed	Silent 504 timeouts	Direct zip download	✓
Mar 6	Watertown not found in TIGER	Named "Watertown Town"	TIGER_NAME_OVERRIDES dict	✓
Mar 6	Cambridge SQMILES = -99 after refresh	STG missing sq miles	UPDATE via ST_AREA()/2589988.11	✓
Mar 7	DBSCAN: StandardScaler destroys lat/long relationship	Geometrically incorrect eps calculation	metric='haversine' on raw radians	✓
Mar 7	Cambridge crime geocoding 10hr runtime	108,726 rows, rate limits	Checkpoint every 200 rows + cache reuses	✓
Mar 8	CA_CRIME_FORCECAST column name mismatch	LOWER_BOUND/UPPER_BOUND vs actual LOWER_CI/UPPER_CI	Updated all references	✓
Apr 9	crime_hotspot_analysis.py requires manual env export	No load_dotenv()	Added load_dotenv()	✓
Apr 14	Geographic neighbors 404 for suburbs	GRANULARITY filter excluded CITY suburbs	Changed to AND CENTROID_LAT IS NOT NULL	✓

dbt / SQL Layer

Session	Issue	Root Cause	Fix	Status
Mar	All spatial-join	Built before	Full refresh of	✓

Session	Issue	Root Cause	Fix	Status
	MARTs had null LOCATION_IDs for Greater Boston	geometry existed	all 8 MRTs	
Mar	KENMORE/BU CENTRAL/LONGWOOD stops in wrong neighborhood	Polygon boundaries generous at edges	Manual override CASE block	✓
Mar	Stop 933 in both Allston AND Brookline	Boundary-straddling stop	ROW_NUMBER() OVER (PARTITION BY stop_id ORDER BY location_id ASC)	✓
Mar	MRT_NEIGHBORHOOD_TRANSIT wrong ref name	Referenced non-existent table	Corrected to MRT_BOSTON_MBTA_STOPS	✓
Mar	MA ZIP codes losing leading zero	Stored as integer	LPAD(TRIM(ZIP_CODE), 5, '0')	✓
Mar	Pho Viet restaurant in Allston AND Brookline	Boundary-straddling restaurant	ROW_NUMBER() dedup	✓
Mar	Universities sources.yml broken — blocked ALL dbt runs	YAML parse error	Fixed YAML to unblock entire pipeline	✓
Apr 14	EXCELLENT grade permanently unreachable (Safety)	Max score 74.0, threshold ≥75	Threshold lowered to ≥65	✓
Apr 14	4 neighborhoods at flat 100 (Transit)	No tiered caps	Tiered caps v3	✓
Feb	LISTAGG + DISTINCT + ORDER BY not	Snowflake limitation	Drop DISTINCT	✓

Session	Issue	Root Cause	Fix	Status
Feb	supported Cortex + c.* causes “Object C does not exist”	Snowflake Cortex limitation	List columns explicitly	✓
Cortex / LLM Layer				
Session	Issue	Root Cause	Fix	Status
Mar 4–7	Arlington→TX, Chelsea→Manha ttan, Everett→WA, Charlestown→“ Charleton”	mistral-7b geo confusion	Upgraded to mistral-large + Massachusetts grounding + REGEXP_REPLACE	✓
Mar	Cortex narrative hallucination for zero-facility healthcare neighborhoods	Cortex generates content even with zero input	Conditional Cortex skip for zero-facility neighborhoods (Rutu)	✓
Mar	Non- deterministic routing on ambiguous queries	cortex_complete at temperature > 0	temperature=0 + keyword overrides + BOTH-biased prompt (Rutu)	✓
Mar	SQL agent used bare table names, mixed- case neighborhoods	Underspecified prompting	Enforced full paths + UPPERCASE in semantic model and prompts (Rutu)	✓
Apr 2	Validator flagging correct rank citations as fabricated	Rank data in Claude’s context but not in validator’s ground truth	Inject rank summaries into verified_detail_metrics	✓
Apr 2	Report Agent silent exit — no PDF generated	Unhandled chart exceptions	All chart generators in try/except, return Optional[str]	✓
Application Layer				
Session	Issue	Root Cause	Fix	Status

Session	Issue	Root Cause	Fix	Status
Mar	embed_batch broke on legal PDFs	Backslashes + non-printable chars in dense regulatory text	Rewrote sanitization: backslashes first → escape quotes → filter non-printable → truncate 2,000 chars (Rutu)	✓
Mar	336K duplicate rows in RAG insert	Row-by-row insert in embed loop	executemany + single conn.commit() (Rutu)	✓
Mar	Duplicate files across domain assignments in RAW_DOMAIN_CHUNKS	Same file assigned to multiple domains	Identified and resolved all cross-domain duplicates (Rutu)	✓
Apr 2	SQL injection in report_agent.py and validator_report_agent.py	f-string interpolation	Parameterized %s queries	✓
Apr 2	Cursor leaks — connections not closed	No try/finally	try/finally with cur.close()	✓
Apr 2	False narrative warning from validator	NARRATIVES key not in data dict	Injected NARRATIVES before Checkpoint 2	✓
Apr 14	“Compare Back Bay and Roxbury” → web_search	compare/vs not in chart keywords	Added to chart keyword list	✓
Apr 14	KeyError: intent in classification prompt	Unescaped JSON braces	Doubled all braces {{ and }}	✓
Apr 14	Graph Agent HTTP calls adding latency	Making HTTP calls	Direct Python import	✓
Apr 14	generate_report name collision	Import and endpoint same	Renamed endpoint	✓

Session	Issue	Root Cause	Fix	Status
Apr 17	in FastAPI Sidebar disappeared	name Browser localStorage	localStorage.c lear() in DevTools + CSS fix	✓
Apr 17	Cost tracker only fires for report queries	Inside wrong if block	Moved outside block	✓
Apr 17	Git merge conflict during live session	Concurrent pushes to main	git config pull.rebase false; git pull origin main	✓

Open / Unresolved Bugs

Priority	Issue	Notes / Workaround
● HIGH	Validation timeout 120s+ per query	Set SKIP_VALIDATION=true env var in handle_data_query()
● HIGH	Grocery 500 error for specific neighborhoods	Column name mismatch in /overview/domain/grocery
● HIGH	KPI cards show neighborhood data when hood_filter active	Stop passing hood_filter to load_kpis()
● MED	Safety donut shows 100% single grade with hood_filter	Show city-wide distribution regardless of filter
● MED	Harbor Islands in CA_CRIME_FORECAST and CA_CRIME_SAFETY_NARRAT IVE	Add HARBOR_FILTER to these queries
● MED	Transit map black background	Get free Mapbox token
● LOW	Python 3.9 Boto3 deprecation	Upgrade to Python 3.10+ before April 29, 2026

18. Current Status Snapshot — April 17, 2026

Data Layer

Component	Status	Details
MASTER_LOCATION	✓ Complete	51 rows, all geometries
All 9 Domain Marts	✓ Complete	51 rows each, Cortex

Component	Status	Details
ANALYTICS.NEIGHBORHOOD_MASTER_SCORE	✓ Complete	narratives Composite weighted score
MRT_BOSTON_CRIME	✓ 356,852+ rows	Boston + Cambridge + Somerville
CA_CRIME_FORECAST	✓ Complete	SARIMAX 6-month per neighborhood
CA_CRIME_HOTSPOT_CLUSTERS	✓ Complete	DBSCAN two-pass
CA_CRIME_SAFETY_NARRATIVE	✓ Complete	Cortex mistral-large
RAW_DOMAIN_CHUNKS	✓ ~8,400 chunks / 70 files	Full corpus (Rutu curated all 8 domains)
Airflow DAGs (crime)	✓ Running	Boston, Cambridge, Somerville weekly
FBI Greater Boston DAG	⚠ Partial	Data through 2024 only
Semantic Model YAML	✓ Complete	20 tables, 19 relationships, uploaded to Snowflake stage
Healthcare DBSCAN	✓ Complete	Proper haversine-based, $\epsilon=250m$

Agent Layer

Agent	Status	Notes
Graph Agent	✓ Complete	100/100 validator pass
Universal Validator	✓ Complete	GPT-4o, first-attempt pass
Router Agent	✓ Complete	3-stage, compare keywords fixed
Cortex SQL Agent	✓ Complete	Queries via semantic model YAML
Web Search Agent	✓ Complete	95/100 on Crime/MBTA
RAG Agent	✓ Complete	8,400 chunks, hybrid scoring
Report Agent v2.1	✓ Complete	Magazine edition PDF, 10 sections
Graphic Agent	✓ Complete	6 chart types + DALL-E 3
Validator Agent	✓ Complete	Pre + post + vision checks

Application Layer

Component	Status	Notes
FastAPI Backend	✓ Complete	All 13 endpoints verified
Input/Output Guardrails	✓ Complete	Wired in FastAPI + Streamlit
LLM Cost Tracker	✓ Complete	Wired (shows 0 for chart queries — known)

Component	Status	Notes
Streamlit Overview Tab	✓ Complete	All 3 modes, Safety + Healthcare deep-dives
Streamlit Ask Tab	✓ Complete	Routing, intent badges, 300s timeout
Streamlit Report Tab	✓ Complete	Poll + PDF download
Docker / EC2 Deployment	✓ Ready	Nginx + HTTPS; t3.large recommended
Project Poster	✓ Final	A4 HTML + 15-slide PPTX (Rutu), QR code to GitHub
System Architecture Diagrams	✓ Final	High-res PNG 2400×2380px (Rutu)

19. Pending Items & Carry-Forward

Priority	Item	Details
● HIGH	Fix validation timeout 120s+	Add SKIP_VALIDATION=true check in <code>handle_data_query()</code>
● HIGH	Fix Grocery 500 error	Debug column names in <code>/overview/domain/grocery</code>
● HIGH	Fix KPI cards always city-wide	<code>load_kpis()</code> must not receive <code>hood_filter</code>
● MED	Build domain deep-dives for 5 remaining domains	Mode 2 + <code>hood_filter</code> for Housing/Transit/Grocery/Schools/Restaurants/Universities
● MED	Transit map with Mapbox token	Get free token, add <code>api_keys={"mapbox": os.getenv("MAPBOX_API_KEY")}</code>
● MED	Fix Harbor Islands in forecast tables	Add HARBOR_FILTER to CA_CRIME_FORECAST and CA_CRIME_SAFETY_NARRATIVE
● MED	Cost tracker for chart queries	Add chart-specific <code>tracker.log()</code> in FastAPI
● LOW	Full Streamlit testing	All 9 domain deep-dives + neighborhood profiles
● LOW	EC2 deployment	Set API_BASE_URL env var, configure Nginx + HTTPS

Priority	Item	Details
 LOW	MBTA + Bluebikes domain agents	Currently handled by graph/router agents
 LOW	LinkedIn post — DBSCAN/SARIMAX	Write-up planned but not done

20. File Reference Index & Commands

File Index

File	Owner	Purpose	Status
neighbourwise_fast_api.py	Aamir	FastAPI backend	
overview_endpoints.py	Rutu	All /overview/* FastAPI endpoints	
router_agent.py	Aamir	3-stage routing pipeline	
neighbourwise_app.py	Aamir	Streamlit frontend	
report_agent.py	Aamir	Report Agent v2.1 Magazine PDF	
graphic_agent.py	Aamir	Chart generation + DALL-E 3	
validator_agent.py	Aamir	Chart + image validator	
validator_report_agent.py	Aamir	Report validator (GPT-4o)	
Graph_agent.py	Aamir	Neo4j LangGraph fan-out	
universal_validator.py	Aamir	Cross-model GPT-4o validator	
neighbourwise_rag.py	Rutu	Production RAG ingest + search	
snowflake_unstructured_loader.py	Yash	Original RAG prototype	
chunk_search.py	Yash	RAG search (hybrid scoring)	
crime_hotspot_analysis.py	Aamir	DBSCAN + SARIMAX + Cortex	
extract_osm_healthcare.py	Rutu	Overpass API healthcare	

File	Owner	Purpose	Status
		extraction	
neighbourwise_guardrails.py	Aamir	Input/output guardrails	✓
LLM_cost_tracker.py	Aamir	Token cost estimation	✓
neighbourwise_semantic_model.yaml	Rutu	20-table semantic model, 19 relationships	✓
upload_yaml.py	Rutu	Uploads semantic model to Snowflake stage	✓
neighbourwise_validator.py	Rutu	validate_and_improve() function	✓
fetch_greater_boston_wkt.py	Aamir	Census TIGER/Line polygon downloader	✓
geocode_boston_crime.py	Aamir	Boston crime geocoder	✓
geocode_cambridge_crime.py	Aamir	Cambridge crime geocoder	✓
geocode_housing.py	Aamir	Housing property geocoder	✓
geocode_grocery.py	Aamir	Grocery store geocoder	✓
neighbourwise_poster_final.html	Aamir	Project poster (A4, QR code)	✓

Key Commands Reference

Airflow

```
cd ~/airflow-project && docker compose up -d
```

```
docker compose logs -f
```

```
docker compose restart airflow-scheduler
```

FastAPI

```
python3 -m uvicorn neighbourwise_fastapi:app --reload --port 8001
```

Streamlit

```
streamlit run neighbourwise_app.py
```

RAG ingest (Rutu's production script)

```
python scripts/neighbourwise_rag.py download --domain GROCERY --url <pdf_url>
```

```
python scripts/neighbourwise_rag.py load --domain GROCERY
```

```
python scripts/neighbourwise_rag.py search --query "grocery access"
```

```
# Crime hotspot analysis
python crime_hotspot_analysis.py

# Report generation test
python3 report_agent.py "Fenway"
python3 report_agent.py "Roxbury"
python3 report_agent.py "Dorchester"

# dbt (CASE-SENSITIVE model names)
dbt run --select MRT_NEIGHBORHOOD_SAFETY
dbt run --select NEIGHBORHOOD_MASTER_SCORE
dbt run --full-refresh --select INT_BOSTON_CRIME

# Rutu's dbt environment (Windows)
deactivate
.\.venv-dbt\Scripts\Activate.ps1
cd dbt\neighborhood_dbt

# Semantic model upload
python scripts/upload_yaml.py

# Git conflict resolution
git config pull.rebase false
git pull origin main
git push origin main

# Clear Streamlit cache (localStorage fix)
# Open DevTools > Console > localStorage.clear() > refresh
```

21. Key Design Decisions & Gotchas

Snowflake account identifier: Use pgb87192 NOT pgb87192.sfedu02 — the .sfedu02 suffix causes SSL cert mismatch with the educational account.

Connector version: Must be snowflake-connector-python==3.6.0 for compatibility with JWT OAuth token authentication.

Sentinel values: -999.0 for lat/long (NOT -99.0 which is a valid longitude value).

INNER JOIN in INT models: Only geocoded records flow downstream — intentional design.

Bounding box geocoding discard: Immediate discard if result outside lat 42.20–42.55, lng -71.35 to -70.85.

LISTAGG + DISTINCT + ORDER BY: Not supported in Snowflake. Drop DISTINCT from LISTAGG expressions.

ST_GEOGRAPHYFROMWKT → VARCHAR: Cannot be cast to VARCHAR — store directly as GEOGRAPHY type.

Cortex + c.*: Causes “Object C does not exist” in Snowflake Cortex. Always list columns explicitly.

--full-refresh in dbt: Required when adding new columns to incremental dbt models.

DIRECTION_ID = 0 in MBTA: Outbound-only filter avoids double-counting routes per neighborhood.

Address deduplication (housing geocoding): Key on ST_NUM+ST_NAME+CITY — geocode once, reuse for all condo units.

NULL zip/city in housing data: Causes AttributeError before .strip() — always null-check first.

source() vs ref(): INT_BOSTON_MBTA_MAPPING is loaded by Airflow → use source(), not ref().

dbt model names CASE-SENSITIVE: Always uppercase: INT_BOSTON_CRIME, MRT_NEIGHBORHOOD_SAFETY etc.

Single-u NEIGHBORHOOD: All Snowflake mart tables use single-u. Teammate dbt refs use double-u mrt_neighbourhood_housing.

MRT_NEIGHBORHOOD_MBTA naming: NOT MRT_NEIGHBORHOOD_TRANSIT.

HARBOR_FILTER: "AND UPPER(NEIGHBORHOOD_NAME) != 'HARBOR ISLANDS'" — missing from CA_CRIME_FORECAST and CA_CRIME_SAFETY_NARRATIVE.

report_agent.py connection: NEVER pass conn from FastAPI. Use generate_report(neighborhood).

Two-step VECTOR INSERT: Snowflake VECTOR type cannot go in VALUES directly. Pattern: temp VARCHAR table → INSERT INTO...SELECT PARSE_JSON()::VECTOR cast.

SARIMAX column names: CA_CRIME_FORECAST uses LOWER_CI and UPPER_CI — **NOT** LOWER_BOUND/UPPER_BOUND.

mistral-large vs mistral-7b: Always use mistral-large. The 7b model hallucinates wrong geographic contexts.

SKIP_VALIDATION env var: Set SKIP_VALIDATION=true in .env and add this check to handle_data_query():

```
if os.environ.get("SKIP_VALIDATION", "false").lower() != "true":
    validated = validate_and_improve(conn, query, draft, sql_data, {"chunks":
rag_chunks})
else:
    validated = {"answer": draft, "improved": False, "feedback": {}}
```

Healthcare 51-neighborhood problem: Never use `HAVING COUNT(...) > 0 + INNER JOIN` against `MASTER_LOCATION` — this silently excludes zero-value neighborhoods. Always use `LEFT JOIN + COALESCE + conditional Cortex skip`.

RAG embed_batch sanitization order matters: Replace backslashes first → then escape single quotes → then filter non-printable/null chars → then truncate to 2,000 chars. Any other order leaves dangerous escape sequences in the text.

dbt environment on Windows (Rutu): Keep `.venv-dbt` (Python 3.12) separate from `.venv`. VS Code auto-activates wrong venv — always deactivate first. A broken `sources.yml` in any domain blocks ALL dbt runs — fix parse errors before running anything.

Cortex Analyst table path format: SQL agent must use fully qualified paths (`NEIGHBOURWISE_DOMAINS.MARTS.TABLE_NAME`) and UPPERCASE neighborhood names. Bare table names and mixed-case cause query failures.

Yelp 1,000-record cap: The Yelp Fusion API caps results at 1,000 per query. Overcome with geographic chunking combined with category-based extraction (Rutu).

*NeighbourWise AI — DAMG 7374 Group 4 — Northeastern University — April 2026
Team: Aamir Jawadwala (Ace) · Yash Khavnekar · Rutu Shah
Compiled from Sessions 1–17 (February–April 2026) including all context logs, dev logs,
Rutu's individual work log, Yash's RAG checkpoints, and early pipeline documentation*